

Home Work 05: Bilinear Rational Quotient Derivative

Course: Mathematics-I (Differential Calculus)

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প্রদত্ত সমীকরণ: $y = \frac{2x + 1}{3x + 1}$

নির্ণেয়: $\frac{dy}{dx}$

ENGLISH NOTE — BILINEAR RATIONAL QUOTIENT FORMULATION

The problem presents a linear fractional transformation of the form $y = \frac{ax+b}{cx+d}$:

$$y = \frac{u(x)}{v(x)}, \quad \text{where } u(x) = 2x + 1, \quad v(x) = 3x + 1$$

Applying the fundamental Quotient Rule:

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}, \quad \text{with } \frac{d}{dx}(2x + 1) = 2, \quad \frac{d}{dx}(3x + 1) = 3$$

১. গাণিতিক কাঠামো ও প্রয়োজনীয় সূত্রসমূহ

উপাদান (Component)	ফাংশন রূপ	প্রমিত অন্তরজ সূত্র (Standard Derivative)
লব (u)	$2x + 1$	$\frac{d}{dx}(2x + 1) = 2$
হর (v)	$3x + 1$	$\frac{d}{dx}(3x + 1) = 3$
হরের বর্গ (v^2)	$(3x + 1)^2$	$(3x + 1)^2$ (উৎপাদক আকারেই রাখা সমীচীন)

ENGLISH NOTE — CRITICAL AMIE EXAM TRAPS & PITFALLS

- Sign Distribution Error:** Be careful when expanding $-(2x + 1)(3) = -(6x + 3) = -6x - 3$. Writing $-6x + 3$ erroneously yields $+5$ in the numerator.
- Factored Denominator:** Keep $(3x + 1)^2$ in factored form rather than expanding into $9x^2 + 6x + 1$.

২. ধাপে ধাপে পূর্ণাঙ্গ সমাধান (Step-by-Step Solution)

ধাপ ১: উভয় পক্ষে x -এর সাপেক্ষে অন্তরীকরণ অপারেটর প্রয়োগ করে:

$$\frac{dy}{dx} = \frac{d}{dx} \left(\frac{2x + 1}{3x + 1} \right)$$

ধাপ ২: ভাগফলের সূত্র (Quotient Rule) অনুসারে সাজানো:

$$\frac{dy}{dx} = \frac{(3x + 1) \frac{d}{dx}(2x + 1) - (2x + 1) \frac{d}{dx}(3x + 1)}{(3x + 1)^2}$$

ধাপ ৩: প্রমিত অন্তরজের মান বসিয়ে:

$$\frac{dy}{dx} = \frac{(3x + 1) \cdot 2 - (2x + 1) \cdot 3}{(3x + 1)^2}$$

ধাপ ৪: লবের বন্ধনী বিস্তার ও পদসমূহ সরলীকরণ:

$$\frac{dy}{dx} = \frac{6x + 2 - (6x + 3)}{(3x + 1)^2} = \frac{6x + 2 - 6x - 3}{(3x + 1)^2} = \frac{-1}{(3x + 1)^2}$$

চূড়ান্ত উত্তর (FINAL STANDARD ANSWER)

$$\frac{dy}{dx} = -\frac{1}{(3x + 1)^2}$$

 ENGLISH NOTE — DOMAIN, MONOTONICITY & ASYMPTOTES

- **Domain of Function & Derivative:** Non-zero denominator requires $3x + 1 \neq 0 \implies x \neq -\frac{1}{3}$. Thus, $\text{Dom}(y) = \text{Dom}\left(\frac{dy}{dx}\right) = \mathbb{R} \setminus \left\{-\frac{1}{3}\right\}$.
- **Strict Decreasing Monotonicity:** Because $(3x + 1)^2 > 0$ for all $x \neq -\frac{1}{3}$ and numerator is negative ($-1 < 0$), $\frac{dy}{dx} < 0$ everywhere on its domain. The curve is strictly decreasing.
- **Asymptotes:** Vertical asymptote at $x = -\frac{1}{3}$; Horizontal asymptote at $y = \lim_{x \rightarrow \pm\infty} \frac{2x+1}{3x+1} = \frac{2}{3}$.

 ENGLISH NOTE — DETERMINANT SHORTCUT FOR BILINEAR FUNCTIONS

For any linear rational function $y = \frac{ax+b}{cx+d}$, the derivative equals:

$$\frac{dy}{dx} = \frac{ad - bc}{(cx + d)^2}$$

Here $a = 2, b = 1, c = 3, d = 1$:

$$ad - bc = (2)(1) - (1)(3) = 2 - 3 = -1 \implies \frac{dy}{dx} = -\frac{1}{(3x + 1)^2}$$

This 5-second determinant shortcut verifies the answer instantly!

 AMIE EXAM MARKS DISTRIBUTION STRATEGY

Exam Evaluation Step	Required Mathematical Rigor	Marks
Quotient Rule Formulation	Correct expansion of $\frac{vu' - uv'}{v^2}$	1.5
Linear Differentiation	Accurate derivatives $\frac{d}{dx}(2x + 1) = 2, \frac{d}{dx}(3x + 1) = 3$	1.5
Algebraic Simplification	Sign expansion $6x + 2 - 6x - 3 = -1$	1.0
Final Standard Format	Writing $-\frac{1}{(3x+1)^2}$	1.0
Total AMIE Section-A Marks		5.0 / 5.0